

AI-Powered Hospital Readmission Prevention Program

Project Management Plan Version 1.0
Healthcare AI Implementation Portfolio Project

1. Project Management Approach

This project will use a hybrid methodology combining predictive project management practices with Agile principles. Healthcare compliance, governance, and approvals require structured planning, while AI development requires iterative testing, feedback, and improvement.

2. Work Breakdown Structure (WBS)

Phase 1: Project Initiation and Planning (Months 1-2)

- Confirm objectives
- Identify stakeholders
- Establish governance structure
- Define success metrics

Phase 2: Data Assessment and Preparation (Months 3-4)

- Identify EHR data sources
- Assess data quality
- Review missing information
- Evaluate bias risks

Phase 3: AI Model Development (Months 5-7)

- Select machine learning approach
- Train model
- Validate performance
- Conduct fairness testing

Phase 4: Clinical Workflow Integration (Months 8-9)

- Define alert process
- Design clinical response workflow
- Train users

Phase 5: Pilot Implementation (Months 10-11)

- Test in selected clinical unit
- Collect feedback
- Evaluate outcomes

Phase 6: Full Deployment (Month 12)

- Hospital rollout
- Monitoring process
- Continuous improvement

3. Project Team Structure

Executive Sponsor: Chief Medical Officer

Project Manager: Healthcare AI Program Manager

Clinical Team: Physicians, nurses, care managers

Technology Team: Data scientists, health IT, cloud engineers

Safety Team: Compliance, ethics, quality improvement

4. Communication Management Plan

Executive leadership meetings: Monthly
AI development meetings: Weekly
Clinical workflow reviews: Biweekly
Risk reviews: Monthly
Stakeholder updates: Throughout project lifecycle

5. Risk Management Plan

Risk: AI bias. Response: Perform fairness testing across patient populations.

Risk: Poor data quality. Response: Validate and clean healthcare data.

Risk: Clinician resistance. Response: Engage stakeholders early and provide training.

Risk: Alert fatigue. Response: Optimize workflow and notification thresholds.

Risk: Privacy concerns. Response: Maintain HIPAA compliance and security controls.

6. Success Metrics

Clinical Metrics:

- Reduction in 30-day readmissions
- Improved follow-up care
- Better patient outcomes

AI Metrics:

- Accuracy
- Sensitivity
- False-positive rates
- Fairness across populations

Operational Metrics:

- Provider adoption
- Response time
- Workflow efficiency

7. Project Deliverables

- Approved project plan
- Data readiness assessment
- Validated AI model
- Clinical workflow design
- Training materials
- Deployment dashboard
- Continuous monitoring plan